8 Logic Families

Introduction

A group of compatible ICs with the same logic levels and supply voltages fabricated for performing various logic functions (OR, AND and NOT) are referred to as logic families.

Classification of logic families:

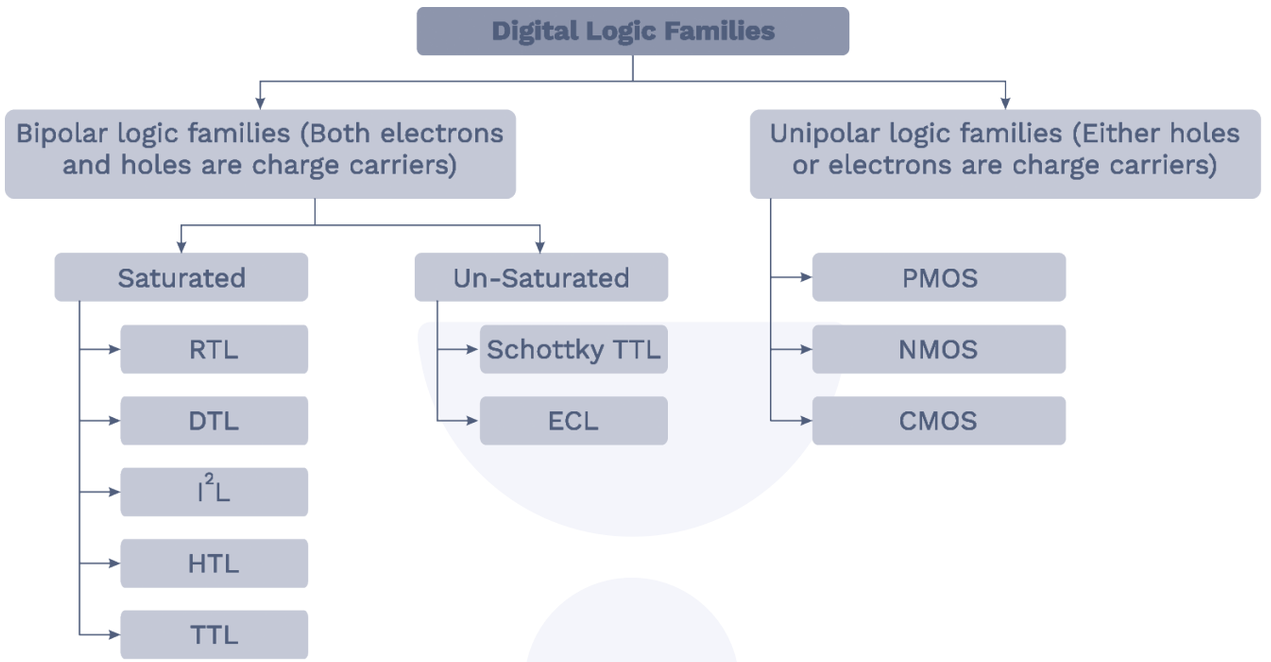


Fig. 8.1 Classification of Logic Families

- Logic circuits which are operated in the saturation region are called saturated logic families.
- Logic circuits that are operated in active and cut-off regions are called unsaturated logic families.

Specifications of logic families:

The important specifications of logic families are:

1. Propagation delay
2. Power dissipation
3. Figure of merit
4. Fan-out
5. Noise margin

1. **Propagation delay:** It is the time required to get a valid output after applying input. It is expressed in nanoseconds. It should be as minimum as possible.
   It depends on the storage time of the transistor and the RC time constant.

Example 1:

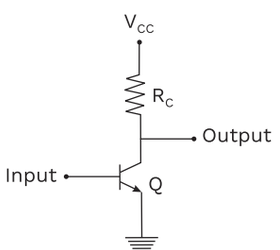


Fig. 8.2

Truth table:

Input	Q	Output
0	OFF	1 (V_{CC})
1	ON	0 (0 Volt)

Table 8.1

This circuit shows the characteristics of an inverter. In this circuit whenever we give the input, the output changes its states from 'ON' to 'OFF' and 'OFF' to 'ON', (i.e., saturation to cut-off and cut off to saturation). So, it requires some time to change its states. Similarly, if some load is applied to this circuit and load contains some capacitance. So, the capacitance requires some time to charge and discharge through this capacitance load. So, based on the switching time of the capacitor and RC time constant, some time is required to get the valid output after applying the input. So, this delay is called propagation delay.

Timing diagram of inverter due to propagation delay:

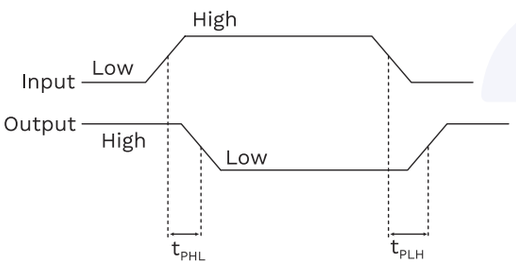


Fig. 8.3 Timing Diagram

In general, propagation delay of the transistor is-

tPLH > tPHL

So, the propagation delay is-

$$t_{pd} = \frac{t_{PHL} + t_{PLH}}{2}$$

2. **Power dissipation:** It is the average power dissipated by one gate. It is expressed in milliwatts. It should be as minimum as possible for any logic gate.

   Power dissipation in transistor

   Pdiss = IC(avg) × VCC

   Where,

$$I_{C(avg)} = \frac{I_{CH} + I_{CL}}{2}$$

3. **Figure of merit:** It is the product of propagation delay (tpd) and power dissipation (Pdiss). It should be as minimum as possible. It is expressed in Pico Joules.

   FOM = tpd ×Pdiss (Pico Joules)

4. **Fan-out:** It is the maximum number of gate inputs we can connect to the output of another gate. It should be as maximum as possible. It depends on the current sinking and current sourcing.

   Current sourcing: When the logic output is in logic '1', the output transistor is in off condition and gate acts like a current source to the external gates connected.

   Current sinking: When the logic output is in logic '0', the output transistor is in on condition and gate acts like a sink to the incoming currents from external gates connected.

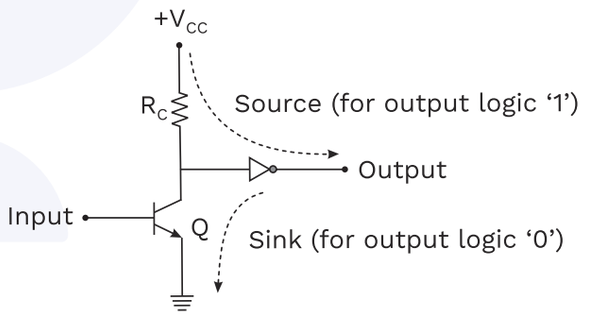


Fig. 8.4 Sinking and Sourcing Current

- When input is '0', the transistor goes to the 'OFF' state, and output will become logic '1'. The current flows from source to the inverter output and the gate acts like a current source to all other external gates connected in output. So the maximum current provided will be-

$$I_{Cmax} = \frac{V_{CC}}{R_C}$$

- When input is '1', transistor goes to the 'ON' state and the output will become logic '0'. So, the gate acts like a sink for all the incoming current of the other external gates connected in the circuit. But here

in this circuit, it cannot sink any current because to stay in saturation region, IC< βIb.

When the output is 'High':

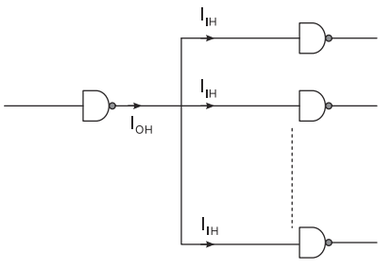


Fig. 8.5 High Output

High state fan-out (FOH) $= \frac{I_{OH}}{I_{IH}}$

When the output is 'Low'

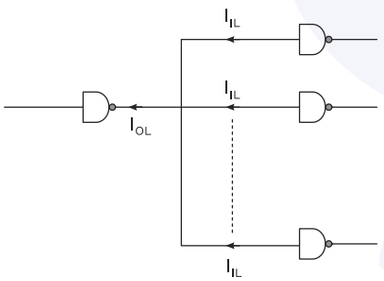


Fig. 8.6 Low Output

Low state fan-out (F_{OL}) $= \frac{I_{OL}}{I_{IL}}$

So, fan-out = min(F_{OH}, F_{OL})

5. Noise margin: It is the maximum amount of noise that can be added at the input of the logic gate, which cannot affect the output logic. It should be as maximum as possible.

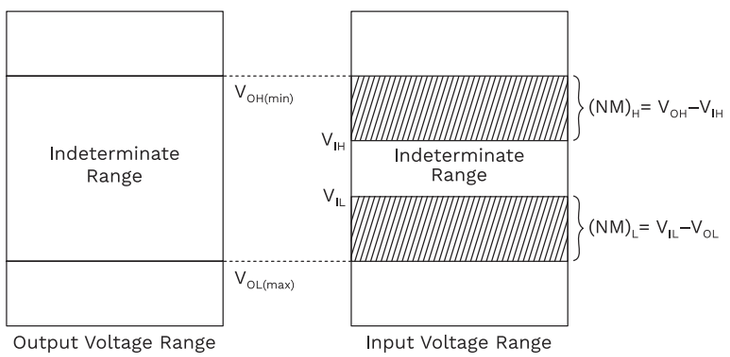


Fig. 8.7 Noise Margin Range

So, $V_{OH} > V_{IH} > V_{IL} > V_{OL}$

Then, noise margin = min $\{(N.M)_H, (N.M)_L\}$

Previous Years' Questions

Given that for a logic family,
V_{OH} is the minimum output high-level voltage.
V_{OL} is the maximum output low-level voltage.
V_{IH} is the minimum acceptable input high-level voltage.
V_{IL} is the maximum acceptable input low-level voltage.
The correct relationship is-
(A) $V_{IH} > V_{OH} > V_{IL} > V_{OL}$
(B) $V_{OH} > V_{IH} > V_{IL} > V_{OL}$
(C) $V_{IH} > V_{OH} > V_{OL} > V_{IL}$
(D) $V_{OH} > V_{IH} > V_{OL} > V_{IL}$
Solution: (B)

Previous Years' Questions

A logic family has threshold voltage $V_R = 2_V$, minimum guaranteed output high voltage V_{OH} = 4V, minimum accepted input high voltage V_{IH} = 3V, maximum guaranteed output low voltage V_{OL} = 1V and maximum accepted input low voltage V_{IL} = 1.5V. Its noise margin is:

(A) 2V (B) 1V
(C) 1.5V (D) 0.5V

Solution: (D)

Example 2:

A logic family has the following specifications:

Minimum guaranteed output high voltage (V_{OH}) = 6V, minimum accepted input high voltage (V_{IH}) = 3V, maximum guaranteed output low voltage (V_{OL}) = 1.5V and maximum accepted input low voltage (V_{IL}) = 2.5V.

The noise margin based on the above specifications will be?

(A) 0.5V (B) 1V
(C) 3V (D) 2.5V

Solution: (B)

$(NM)_H = V_{OH} - V_{IH} = 6 - 3 = 3V$

$(NM)_L = V_{IL} - V_{OL} = 2.5 - 1.5 = 1V$

So, noise margin = $\min\{(NM)_H, (NM)_L\}$

Noise margin = $\min\{3, 1\}$

Noise margin = 1V

Correct answer is (B).

Previous Years' Questions

The inverter 74ALS01 has the following specifications:

$I_{OHmax} = -0.4$ mA, $I_{OLmax} = 8$ mA
$I_{IHmax} = 20$ μA, $I_{ILmax} = -0.1$ mA

The fan-out based on the above data will be:

(A) 10 (B) 20
(C) 60 (D) 100

Solution:

Resistor Transistor Logic (RTL)

Transistor is called as a current-controlled device, it means that the base current (i_b) in the transistor controls the transistor.

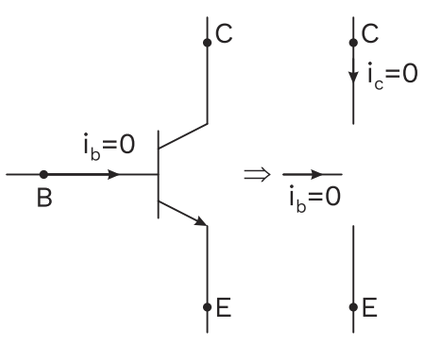


Fig. 8.8 RTL

In transistor, if the base current i_b = 0, then the transistor acts like an open circuit. So, no current will flow from collector to emitter. If $i_b \neq 0$, then the transistor goes into the ON-state. So, the transistor acts like a short circuit, and the collector current (i_c) flows to the emitter terminal.

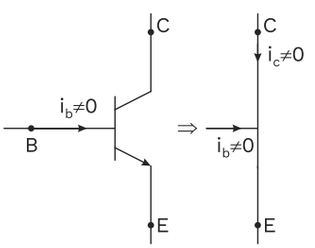


Fig. 8.9 Collector to Emitter

If i_b = 0, then the transistor is in a cut-off state, and no current flows from collector to emitter.

If $i_b \neq 0$, then the transistor goes into the ON-state, and current (i_c) flows from collector to emitter.

Inverter:

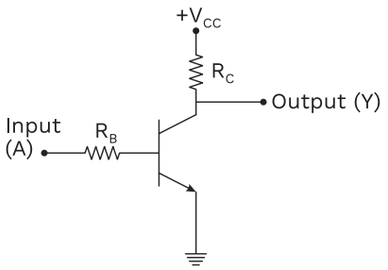


Fig. 8.10 Inverter Mode

In the given circuit, if input = 0 (A = low (or) logic '0') then no current flows from the base terminal (i_b= 0). So, the transistor acts like an open circuit and supply voltage +V_{CC} appears at the output terminal (output = V_{CC} (or) Y = logic '1').

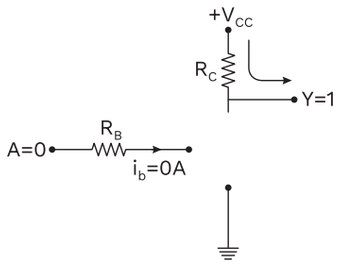


Fig. 8.11 Open Circuited

If input = 1 (logic 'high' (or) A = 1), then the base current $i_b \neq$ 0 and transistor act like a short circuit and the logic 'low' appears at the output terminal (Y = logic '0').

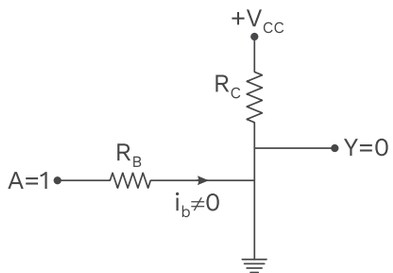


Fig. 8.12 Short Circuited

Truth Table:

Input	Output
A	**Y**
0 (Low)	1 (High)
1 (High)	0 (Low)

Table 8.2

The above truth table shows the inverter operation. So, the given circuit is an inverter or NOT gate.

Buffer:

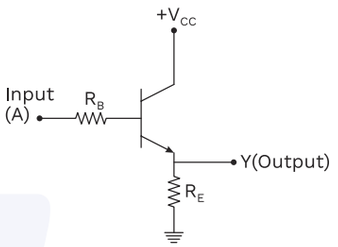


Fig. 8.13 Buffer

In the given circuit, if input (A) is logic 'low' (or, A = 0), then the base current (i_b = 0) is zero. So, the transistor acts like an open circuit and the output goes into the low state (Y = logic '0').

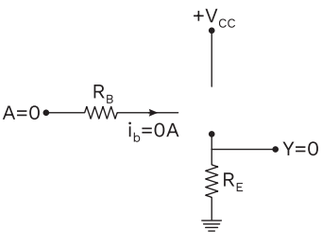


Fig. 8.14 Open Circuited

If input (A) is logic 'high' (or, A = 1), then the base current ($i_b \neq$ 0) flows through the transistor. So, the transistor goes into the ON-state and output becomes logic 'high' (or, Y = 1).

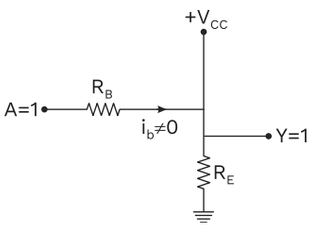


Fig. 8.15 Short Circuited

Previous Years' Questions

The logic functionality realised by the circuit shown below is

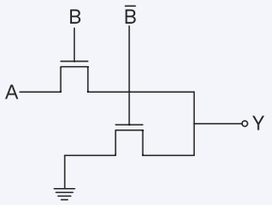


Fig. 8.16

(A) OR (B) XOR
(C) NAND (D) AND
Solution: (D)

Truth table:

Input	Output
A	Y
0 (Low)	0 (Low)
1 (High)	1 (High)

Table 8.3

The above truth table shows the buffer operation. So, the given circuit is a buffer circuit.

NAND gate:

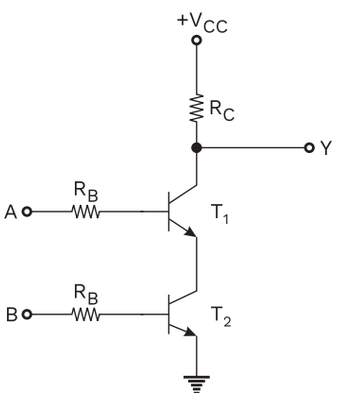


Fig. 8.17 NAND Gate

In the above circuit, if both input A and B are 'low' (logic '0'), then both transistors T1 and T2 go into OFF-state (open circuit). So, the output (Y) becomes 'high' (logic '1').

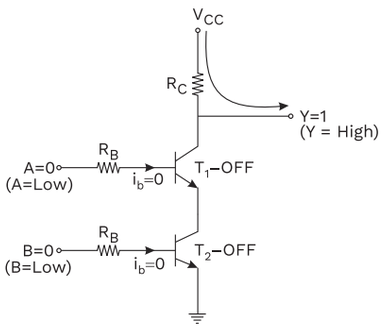


Fig. 8.18 For both Input Low

If input A is low (logic '0') and B is high (logic '1'), then only the transistor T2 goes into ON-state. So, the output (Y) becomes 'high' (logic '1').

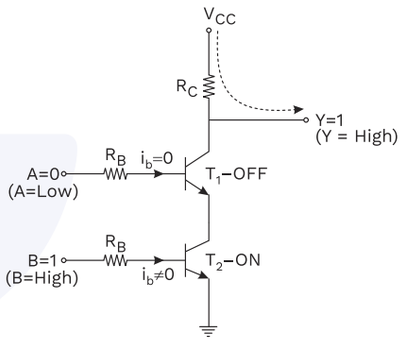


Fig. 8.19 For One Input High and One Input Low

If input A is high (logic '1') and B is low (logic '0'), then the transistor T1 goes into the ON-state. So, the output (Y) becomes 'high' (logic '1').

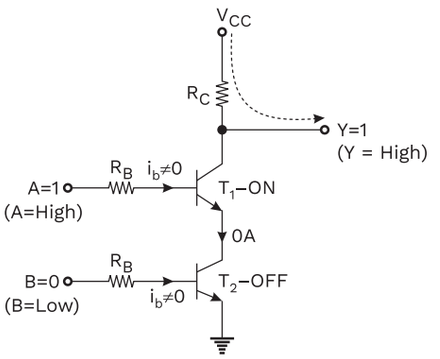


Fig. 8.20 For One Input High and One Input Low

If input A is high (logic '1') and input B is high (logic '1'), then the transistors T1 and T2 both go into ON-state. So, the output (Y) becomes 'low' (logic '0').

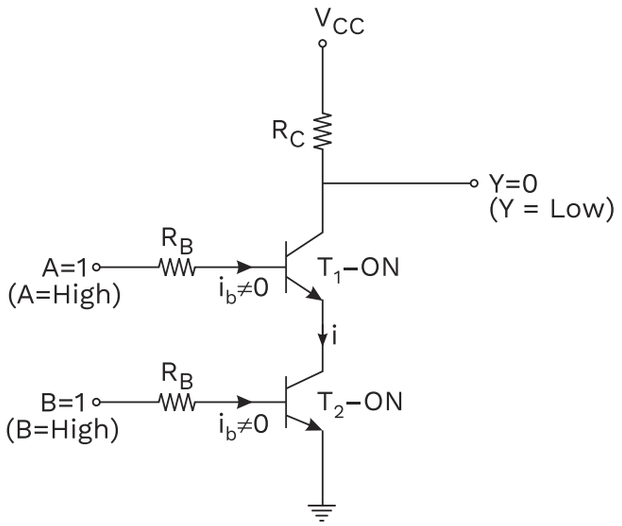


Fig. 8.21 For Both Input High

Truth table:

Input		Output
A	**B**	**Y**
Low (0)	Low (0)	High (1)
Low (0)	High (1)	High (1)
High (1)	Low (0)	High (1)
High (1)	High (1)	Low (0)

Table 8.4

The above truth table shows the NAND operation. So, the given circuit is NAND gate.

NOR gate:

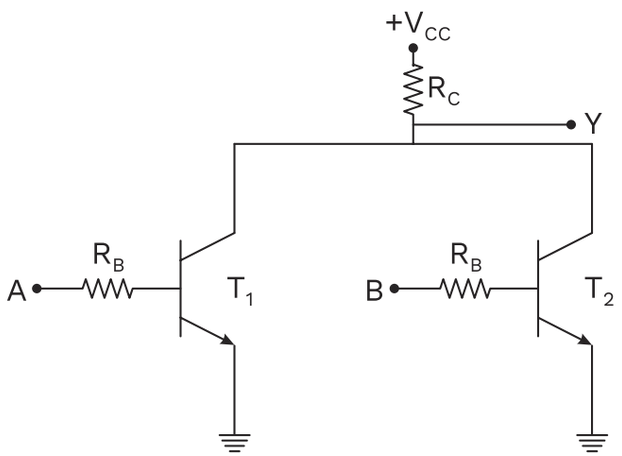


Fig. 8.22 NOR Gate

- If input A = 0 (low) and B = 0 (low), then both the transistors T_1 and T_2 go into the OFF-state. So, the output becomes 'high' (Y = 1).
- If input A = 0 (low) and B = 1 (high), then the transistor T_1 goes into the OFF-state and transistor T_2 goes into the ON-state. So, the output becomes 'low' (Y = 0).
- If input A = 1 (high) and B = 0 (low), then the transistor T_1 goes into the ON-state and transistor T_2 goes into the OFF-state. So, the output becomes 'low' (Y = 0).
- If input A = 1 (high) and B = 1 (high), then both the transistors go into the ON-state and the output becomes 'low' (Y=0).

Previous Years' Questions

In the circuit shown, what are the values of F for EN = 0 and EN = 1, respectively?

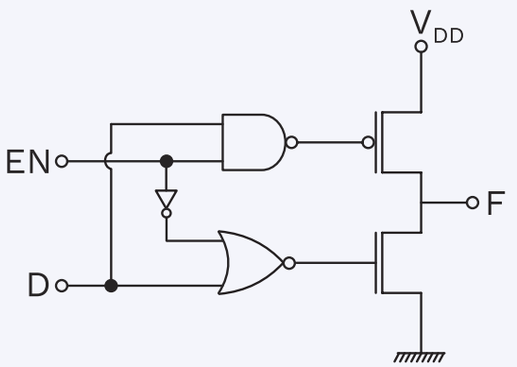


Fig. 8.23

(A) 0 and 1
(B) Hi-Z and $\underline{D}$
(C) Hi-Z and $\overline{D}$
(D) 0 and D
Solution: (B)

Truth table:

Input		Output
A	**B**	**Y**
Low (0)	Low (0)	High (1)
Low (0)	High (1)	Low (0)

Input		Output
A	B	Y
High (1)	Low (0)	Low (0)
High (1)	High (1)	Low (0)

Table 8.5

The above truth table shows the NOR operation. So, the given circuit is NOR circuit.

Example 3:

In the given circuit, what will be the expression for Y? Which gate is realised by the given circuit?

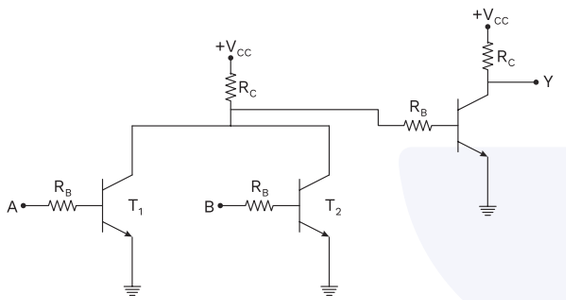


Fig. 8.24

Solution:

Given circuit,

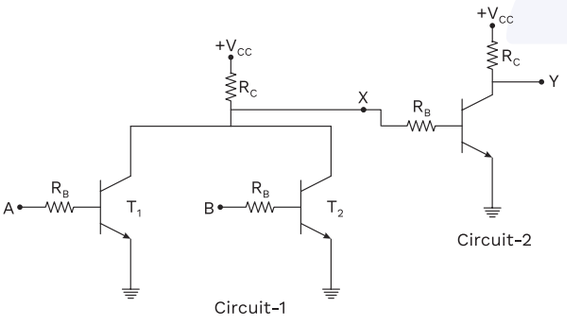


Fig. 8.25

Let X be the output of circuit-1 and Y be the output of circuit-2.

The given circuit-2 is an inverter circuit. So

$$Y = \overline{X}$$

Truth table:

Input		Output	
A	B	X	$Y = \overline{X}$
0	0	1	0
0	1	0	1
1	0	0	1
1	1	0	1

Table 8.6

So, $X = \overline{A}\,\overline{B}$

$$X = \overline{A + B}$$

Output, $Y = \overline{X}$

$$Y = \overline{\overline{A + B}}$$

$$Y = (A+B)$$

Hence, the given circuit is OR gate and expression Y = (A+B).

Specifications of RTL:

- RTL supports wired AND operation
- Propagation relay (t_{pd}) = 50 nsec
- Power dissipation (P_{diss}) = 10 mW
- Figure of merit (FOM) = 500 pJ
- Noise margin (NM) = 0.2 V
- Fan-out = 3

Logic Gates Implementation Using Diodes:

In an ideal diode, if the anode voltage (VA) is more than the cathode voltage (VC), then the diode is in forward bias and acts like a short circuit.

If the anode voltage (VA) is less than the cathode voltage (VC), then the diode is in reverse bias and acts like an open circuit.

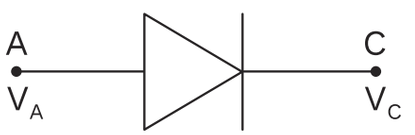


Fig. 8.26 Diode

If VA>VC, diode is in forward bias (short circuit)

If VA<VC, diode is in reverse bias (open circuit)

Previous Years' Questions

In the circuit shown below, Q1 has negligible collector-to-emitter saturation voltage, and the diode drops negligible voltage across it under forward bias. If V_{CC} is +5V, X and Y are digital signals with 0V as logic 0 and V_{CC} as logic 1, then the Boolean expression for Z is:

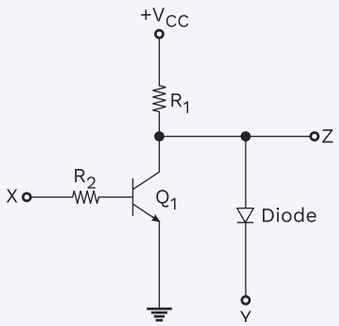


Fig. 8.27

(A) XY (B) $\overline{XY}$

(C) $X\overline{Y}$ (D) $\overline{X}\,\overline{Y}$

Solution: (B)

AND gate using diodes:

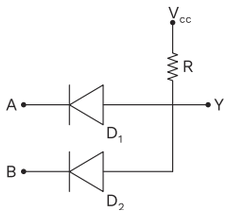


Fig. 8.28 AND Gate Using Diodes

- The given circuit is of AND gate. If inputs applied are A = 0 and B = 0, then both the diodes D1 and D2 become forward bias. So, the diodes act as short circuits and the output of the circuit becomes logic '0' (or, Y = 0).

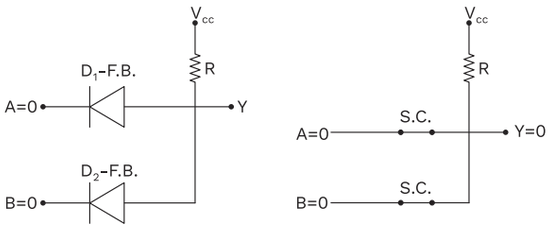


Fig. 8.29 Both Diode Forward Biased

- If inputs applied are A = 0 and B = 1, then diode D1 becomes forward bias (short circuit) and D2 becomes reverse bias (open circuit), and the output of the circuit becomes logic '0' (or, Y = 0).

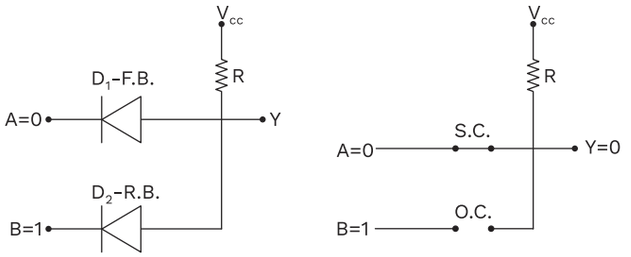


Fig. 8.30 One Diode Forward Biased and One Diode Reverse Biased.

- If inputs applied are A = 1 and B = 0, then diode D1 becomes reverse bias (open circuit) and D2 becomes forward bias (short circuit) and the output of the circuit becomes logic '0' (or, Y = 0).

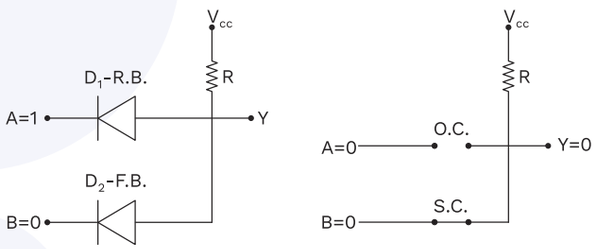


Fig. 8.31 One Diode Forward Biased and One Diode Reverse Biased.

- If inputs applied are A = 1 and B = 1, then both the diodes D1 and D2 become open circuit (reverse bias). So the output of the circuit becomes logic '1' (or, Y = 1).

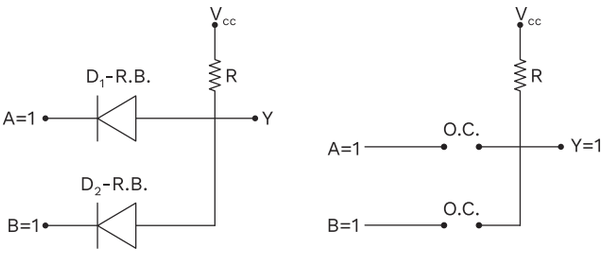


Fig. 8.32 Both Diodes are Reverse Biased

Truth table:

Inputs		Output
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Table 8.7

Previous Years' Questions

In the circuit shown, diodes D_1, D_2 and D_3 are ideal, and the inputs E_1, E_2 and E_3 are '0 V' for logic '0' and '10 V' for logic '1'. Which logic gate does the circuit represent?

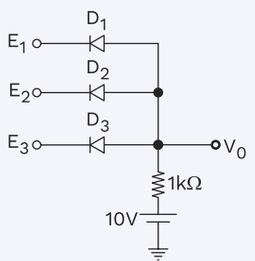


Fig. 8.33

(A) 3-input OR gate
(B) 3-input NOR gate
(C) 3-input AND gate
(D) 3-input XOR gate
Solution: (C)

Example 4:

Specify the operation of the circuit having inputs A, B and C, and output Y.

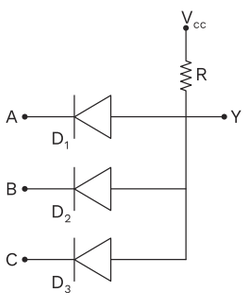


Fig. 8.34

Solution:

Truth table:

A	B	C	D_1	D_2	D_3	Y
0	0	0	Forward bias	Forward bias	Forward bias	0
0	0	1	Forward bias	Forward bias	Reverse bias	0
0	1	0	Forward bias	Reverse bias	Forward bias	0
0	1	1	Forward bias	Reverse bias	Reverse bias	0
1	0	0	Reverse bias	Forward bias	Forward bias	0
1	0	1	Reverse bias	Forward bias	Reverse bias	0
1	1	0	Reverse bias	Reverse bias	Forward bias	0
1	1	1	Reverse bias	Reverse bias	Reverse bias	1

Table 8.8

So, the output of the circuit Y = ABC.

OR Gate using diodes:

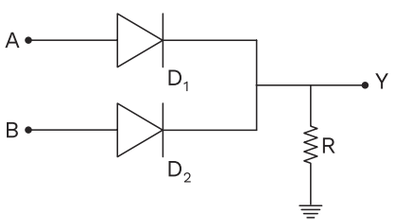


Fig. 8.35 OR Gate Using Diodes

- If inputs applied are A = 0 and B = 0, then both the diodes D1 and D2 become reverse bias (open circuit), and the output of the circuit becomes logic '0' (or, Y=0).

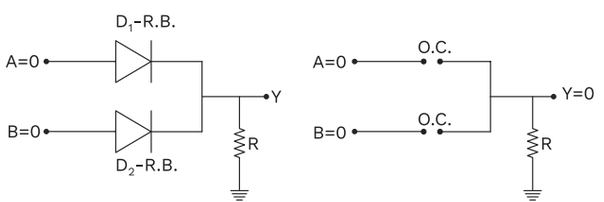


Fig. 8.36 OR Gate Using Diodes

- If inputs applied are A = 0 and B = 1, then diode D_1 becomes reverse bias (open circuit) and diode D_2 becomes forward bias (short circuit), and the output of the circuit becomes logic '1' (or, Y = 1).

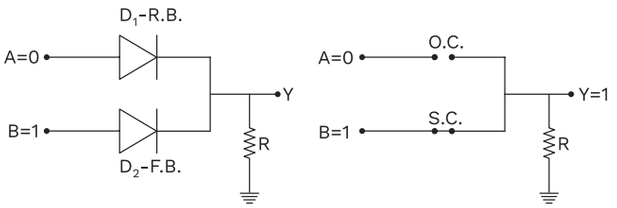


Fig. 8.37 OR Gate Using Diodes

- If inputs applied are A = 1 and B = 0, then diode D_1 becomes forward bias (short circuit) and the diode D_2 becomes reverse bias (open circuit) and the output of the circuit becomes logic '1' (or, Y = 1).

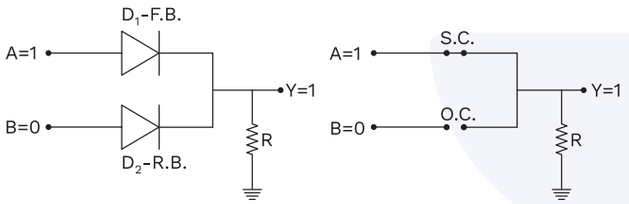


Fig. 8.38 OR Gate Using Diodes

- If inputs applied are A = 1 and B = 1, then both the diodes D_1 and D_2 become forward bias (short circuit), and the output of the circuit becomes logic '1' (or, Y = 1).

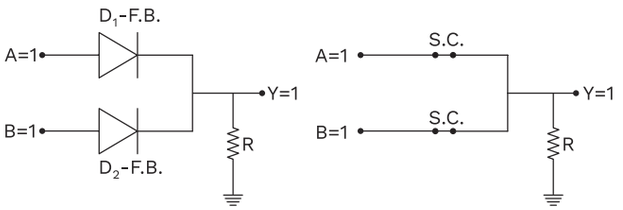


Fig. 8.39 OR Gate Using Diodes

Truth table:

Inputs		Output
A	**B**	**Y**
0	0	0
0	1	1
1	0	1
1	1	1

Table 8.9

Example 5:

Find the expression for Y as shown in the figure.

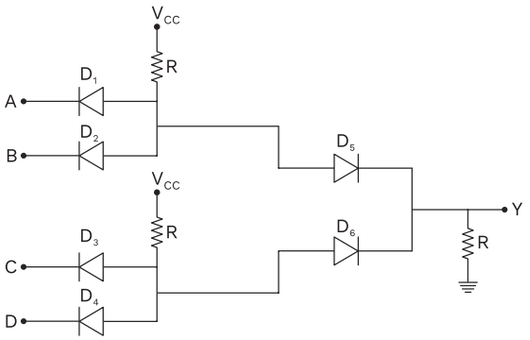


Fig. 8.40

Solution:

Given circuit,

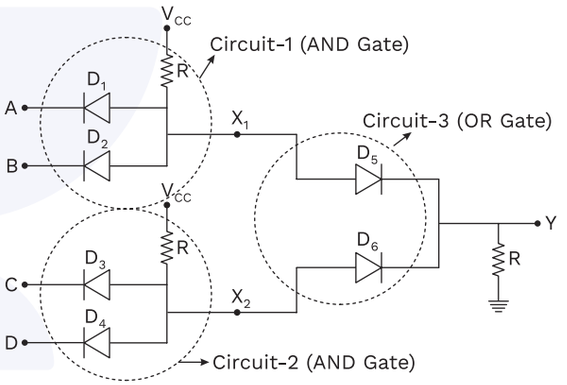


Fig. 8.41

Previous Years' Questions

Among the digital IC-families, ECL, TTL and CMOS:

(A) ECL has the least propagation delay

(B) TTL has the largest fan-out

(C) CMOS has the biggest noise margin

(D) TTL has the least power consumption

Solution: (A)

The given circuit is a combination of three circuits, where circuit-1 and circuit-2 are AND gate, and circuit-3 is the diode representation of OR Gate.

Output of circuit-1 → $X_1 = AB$

Output of circuit-2 → $X_2 = CD$

Output of circuit-3 → $Y = X_1 + X_2$

$Y = AB + CD$

Transistor Transistor Logic (TTL)

The transistor logic circuit uses transistors to perform logic functions. TTL logic circuits have three different configurations-

1. Open collector output TTL
2. Tri-state output TTL
3. Totem pole output TTL

1. Open collector output TTL:

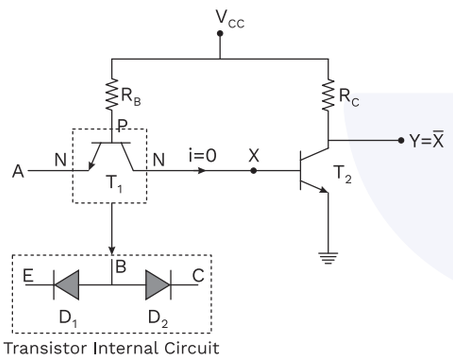


Fig. 8.42 Open Collector Output TTL

In the above TTL circuit, if we take input A=0, then the emitter terminal of the transistor (T_1) becomes logic '0' (or 0 volt) and the base voltage is 'high'. So, diode D1 becomes forward bias and diode D_2 becomes reverse bias. Current flows from base to emitter terminal and no current flows from base to collector terminal (D_2 is reverse bias). So, the transistor (T_2) becomes OFF (base current of T_2 is ib = 0) and the output of the TTL circuit will become logic '1' (or, Y = 1).

If input A = 1, then diode D1 becomes reverse bias and diode D2 becomes forward bias. So, the transistor (T_2) goes into the ON-state and the output of the TTL circuit becomes logic '0' (or Y = 0).

Truth table:

Input	Output
A	Y
0	1
1	0

Table 8.10

Previous Years' Questions

The figure shows the internal schematic of a TTL AND-OR-Invert (AOI) gate. For the inputs shown in the figure, the output Y is

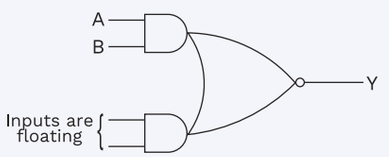


Fig. 8.43

(A) 0 (B) $\overline{1}$

(C) AB (D) $\overline{AB}$

Solution: (A)

In the TTL circuit, if suddenly some transient occurs in the network, then the base current becomes very high, which can damage the transistor (T_1). So, to eliminate the drawback of this circuit, we use a diode (D) at the emitter terminal of transistor (T_1).

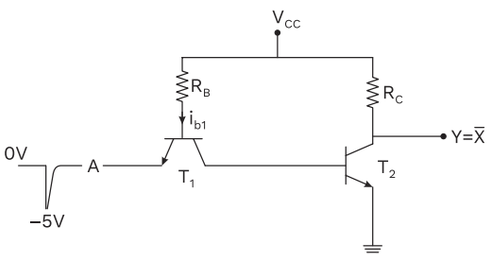


Fig. 8.44

$$i_{b_1} = \frac{V_{CC} - [-5]}{R_B}$$

$$i_{b_1} = \frac{V_{CC} + 5}{R_B}$$

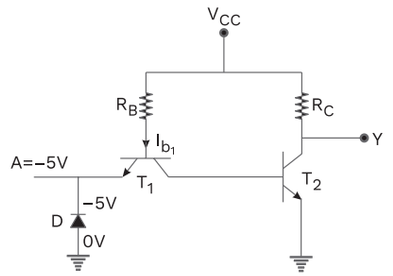


Fig. 8.45

Base current of the transistor should be low (in order of mA), but here the base current (ib_1) is very high.

a) When –ve spikes occurs:
Diode (D) becomes short circuit and emitter terminal gets connected to the ground (0 V).

So, Base current $\left(i_{b_1}\right) = \frac{V_{CC} - 0}{R_B}$

b) When +ve spikes occurs:
Diode (D) becomes open circuit and emitter terminal voltage of transistor T_1 will become +5 volt.

So, Base current $\left(i_{b_1}\right) = \frac{V_{CC} - (-5)}{R_B}$

c) Diode (D) is called as a clamping diode.

2. Tri-state output TTL:
Tri-state logic provides high-speed operation with wired logic. It has three states: low, high and high impedance state.

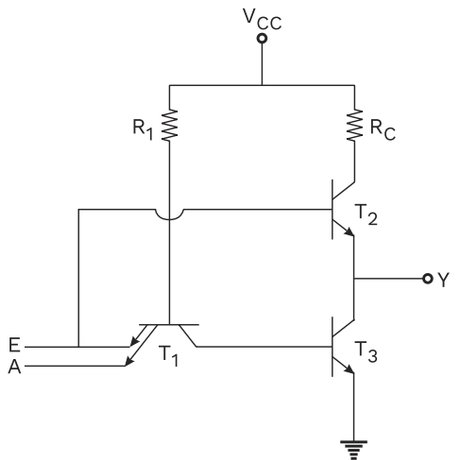


Fig. 8.46 Tri-State Output TTL

a) In a tri-state logic circuit, if E = 0 and A = 0, then transistor T2 will become OFF (open circuit) and T1 will become ON (short circuit), transistor T3 will also become OFF (base current = 0). So, the output of the tri-state circuit goes into the 'High impedance state'.
b) If E = 1 and A = 0, then transistor T2 will be in ON-state, T1 will be in OFF-state (reverse bias) and T3 will be in OFF-state. So, the output of the tri-state circuit becomes logic '1' (or Y = 1).
c) If E = 1 and A = 1, then transistor T2 will be in ON-state, T1 will be in ON-state and T3 will be in ON-state. So, the output of the tri-state circuit becomes logic '0' (or Y = 0).

Truth table:

E	A	Y
0	X	High-impedance state
1	0	1
1	1	0

Table 8.11

3. Totem pole output/active pull-up stage TTL:

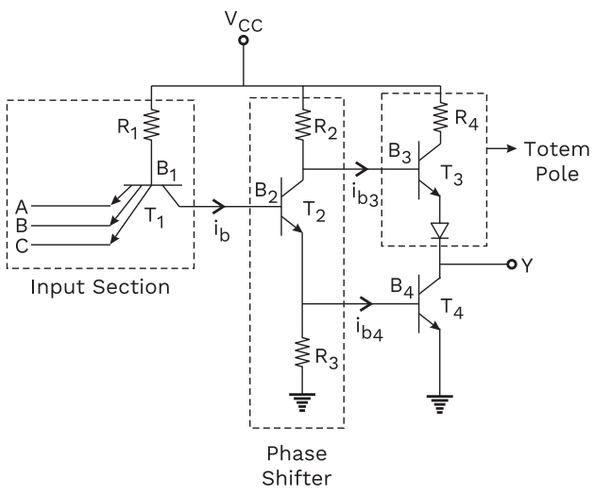


Fig. 8.47 Totel Pole Output TTL

Totem pole configuration of the TTL circuit consists of three sections. Transistor T_1 is included in the input section, transistor T_2

is included in the phase shifter section and transistor T_3 is included in totem pole section (active pull-up element).

Input section:

In input section of the totem pole configuration, there are three emitter terminals A, B and C. So, we can replace every base to emitter junction with diodes D_1, D_2 and D_3.

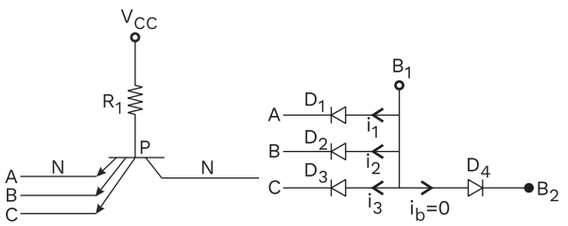


Fig. 8.48

1. If any one of the inputs (A, B and C) is logic '0' (low), then the corresponding diodes (D_1 or D_2 or D_3) act as forward bias (short circuit), and no current will flow through diode D_4 (i_b=0).
2. If all the three inputs (A = B = C = 1) are logic '1' (high), then all the three diodes (D_1, D_2 and D_3) act as reverse bias (open circuit), and current will flow through the diode D_4 ($i_b \neq 0$).

Output section (phase shifter + totem pole):

The output section of the totem pole configuration takes the current from the input section, and that current (i_b) becomes the base current for the transistor (T_2).

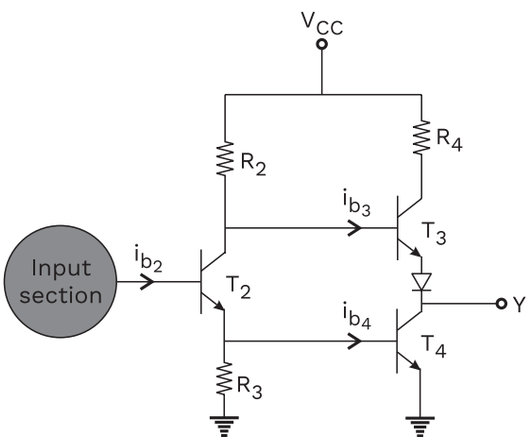


Fig. 8.49

1. If base current of transistor (T_2), i.e., i_{b2} = 0, then the transistors T_2 and T4 goes into the OFF-state and the transistor (T_3) goes into the ON-state (base current, $i_{b3} \neq 0$). So, the output (Y) becomes logic '1' (or, Y = 1).
2. If base current of transistor (T_2), i.e., $i_{b2} \neq$ 0, then the transistors T_2 and T_4 go into the ON-state and transistor (T_3) goes into the OFF-state (base current, i_{b3} = 0). So, the output (Y) becomes logic '0' (or Y = 0) for input A = B = C = 1.

Truth table:

A	B	C	T_1	T_2	T_3	T_4	Y (Output)
0	0	0	ON	OFF	ON	OFF	1 (High)
0	0	1	ON	OFF	ON	OFF	1 (High)
0	1	0	ON	OFF	ON	OFF	1 (High)
0	1	1	ON	OFF	ON	OFF	1 (High)
1	0	0	ON	OFF	ON	OFF	1 (High)
1	0	1	ON	OFF	ON	OFF	1 (High)
1	1	0	ON	OFF	ON	OFF	1 (High)
1	1	1	OFF	ON	OFF	ON	0 (Low)

Table 8.12

Advantages of totem pole configuration:

- Higher speed of operation
- Higher fan-out
- Low power dissipation

Specifications of TTL:

- Propagation delay (t_{pd}) = 10 nsec
- Power dissipation (P_{diss}) = 10 mW
- Fan-out = 10
- Figure of merit (FOM) = 100 pJ
- Noise margin (NM) = 0.4 V

Emitter Coupled Logic (ECL)

- ECL transistors are operated between cutoff and active region. Hence, it is a non-saturated logic family.
- Fastest among all the logic families.

- In ECL, RTL and DCTL, floating inputs will act as logic '0'.
- ECL logic family provides wired OR operation.
- ECL works on negative logic system.

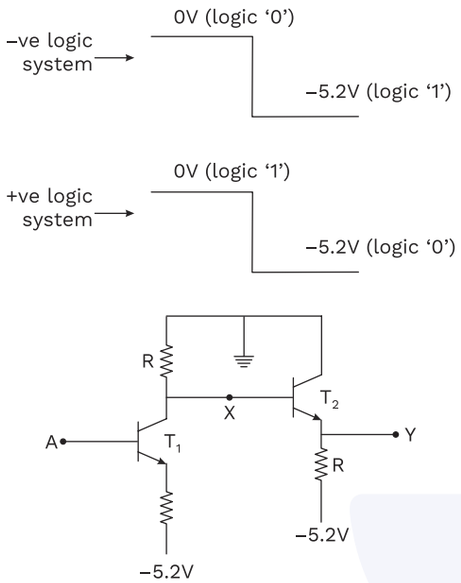


Fig. 8.50 ECL

Previous Years' Questions

In the circuit shown, A and B are the inputs and F is the output. What is the functionality of the circuit?

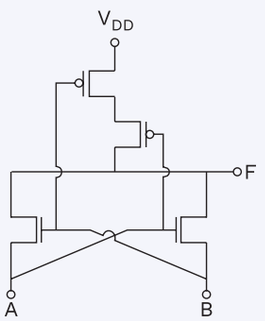


Fig. 8.51

(A) X NOR (B) XOR
(C) Latch (D) SRAM Cell
Solution: (A)

- If input A = 0 (logic '0'), then the base-emitter junction of transistor T_1 becomes forward bias (short circuit), and base-collector junction of transistor T_1 becomes reverse bias (open circuit). So, the output of T_1 becomes zero (X = 0), which makes the transistor T_2 OFF (open circuit), and the output of the circuit becomes logic '1' (Y = –5.2V).
- If input A = 1 (logic '1'), then the base-emitter junction of transistor T1 becomes forward bias (short circuit) and base-collector junction of transistor T_1 becomes forward bias (short circuit). So, the output of T_1 becomes '1' (X = 1), which makes the transistor T_2 ON (short circuit), and the output of the circuit Y becomes logic '0' (Y = 0-V).

Truth table:

Input	Output
A	Y
0	1
1	0

Table 8.13

- ECL uses negative power supply. Due to this power supply ripples (or) spikes will not affect the operation.
- In ECL, common collector output is used to improve fan-out.

Specifications of ECL

- Propagation delay (t_{pd}) = 1n sec
- Power dissipation (P_{diss}) = 55 mW
- Figure of merit (FOM) = 55 pJ
- Fan-out = 25
- Noise margin (NM) = 0.3 V

Mos Logic Families

MOS logic family is easy to fabricate and occupies a very small space. It does not require resistors and diodes. Because of its ease of fabrication and lower power dissipation, it is widely used in LSI, VLSI and ULSI. Because of the very high impedance present at MOSFET's input, the MOS logic families are more susceptible to static charge damage.

The MOSFETs are of two types:

1. NMOS (N-Type MOSFET)
2. PMOS (P-Type MOSFET)
   a) The speed of NMOS is three times that of PMOS. So, NMOS is widely used in fabrication.
   b) Symbols of NMOS

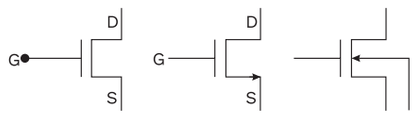


Fig. 8.52 MOSFET Symbol

VGS > 0 → NMOS is in ON-state → short circuit

VGS < 0 → NMOS is in OFF-state → open circuit

   c) In MOSFET, if gate and drain are shorted, then MOSFET behaves as a resistor.

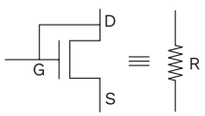


Fig. 8.53

NMOS NOT gate:

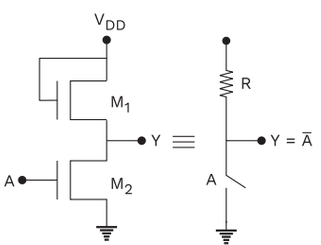


Fig. 8.54 NMOS NOT Gate

- If A = 0, then the MOSFET (M_2) will be in OFF-state, and the output of NMOS circuit will be logic '1' (Y = 1).
- If A = 1, then the MOSFET (M_2) will be in ON-state, and the output of NMOS circuit will be logic '0' (Y = 0).

Truth table:

A	M_2	Y
0	OFF	1
1	ON	0

Table 8.14

Previous Years' Questions

In the figure, Boolean expression for the output in terms of inputs A, B and C when the clock 'Ck' is high, is:

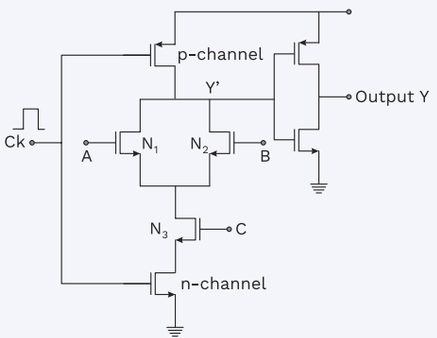


Fig. 8.55

(A) A(B+C) (B) C(B+A)
(C) B(A+C) (D) ABC

Solution: (B)

NMOS NAND Gate

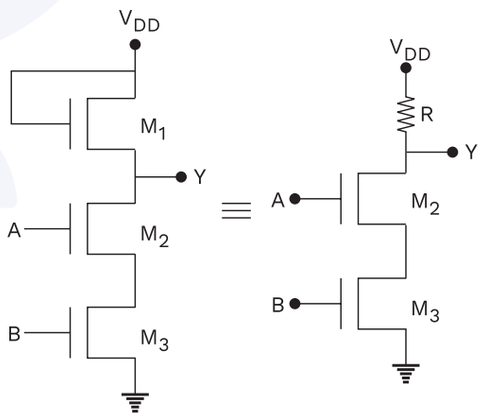


Fig. 8.56 NMOS NAND Gate

- If A = 0 and B = 0, then the MOSFETs M_2 and M_3 will be in OFF-state, and the output of NMOS circuit will be logic '1' (Y = 1).
- If A = 0 and B = 1, then the MOSFET M_2 will be in OFF-state and M_3 will be in ON-state and the output of NMOS circuit will be logic '1' (Y = 1).
- If A = 1 and B = 0, then the MOSFET M_2 will be in ON-state and M_3 will be in OFF-state and the output of NMOS circuit will be logic '1' (Y = 1).

- If A = 1 and B = 1, then the MOSFETs M_2 and M_3 will be ON-state and the output of NMOS circuit will be logic '0' (Y = 0).

Previous Years' Questions

For the NMOS logic gate shown in figure, the logic function implemented is:

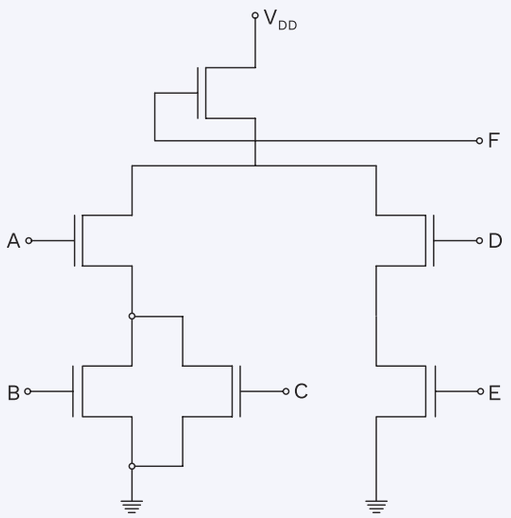


Fig. 8.57

(A) $\overline{ABCDE}$

(B) $(AB+\overline{C})\overline{(D+E)}$

(C) $\overline{A(B+C)+DE}$

(D) $\overline{(A+B)}C+\overline{D}\overline{E}$

Solution: (C)

Truth table:

A	B	M_2	M_3	Y
0	0	OFF	OFF	1
0	1	OFF	ON	1
1	0	ON	OFF	1
1	1	ON	ON	0

Table 8.15

NMOS NOR GATE

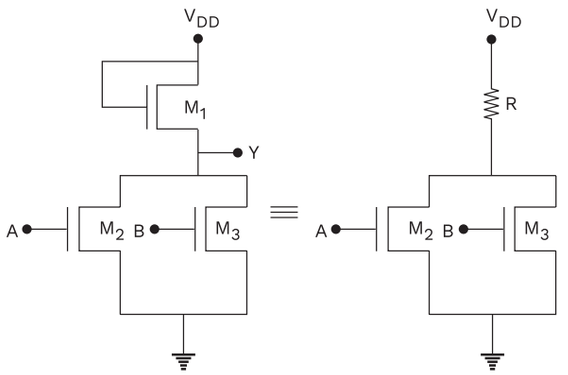


Fig. 8.58 NMOS NOR Gate

- If A = 0 and B = 0, then the MOSFETs M_2 and M_3 will be in OFF-state, and the output of NMOS circuit will be logic '1' (Y = 1).
- If A = 0 and B = 1, then the MOSFET M_2 will be in OFF-state, and M_3 will be in ON-state, and the output of NMOS circuit will be logic '0' (Y = 0).
- If A = 1 and B = 0, then the MOSFET M_2 will be in ON-state and M_3 will be in OFF-state and the output of NMOS circuit will be logic '0' (Y = 0).
- If A = 1 and B = 1, then the MOSFETs M_2 and M_3 will be in ON-state and the output of NMOS circuit will be logic '0' (Y = 0).

Truth table:

A	B	M_2	M_3	Y
0	0	OFF	OFF	1
0	1	OFF	ON	0
1	0	ON	OFF	0
1	1	ON	ON	0

Table 8.16

Previous Years' Questions

The output (Y) of the circuit shown in the figure is:

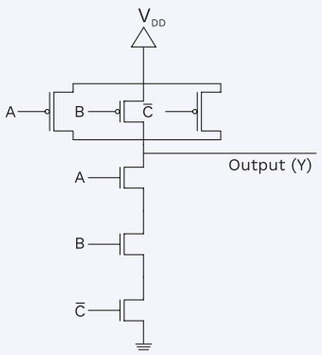


Fig. 8.59

(A) $\bar{A} + \bar{B} + C$

(B) $A + \bar{B}\bar{C} + A\bar{C}$

(C) $\bar{A} + B + \bar{C}$

(D) $AB\bar{C}$

Solution: (A)

Example 5:

Find the expression for Y in the given MOSFET circuit.

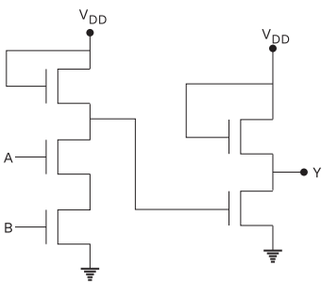


Fig. 8.60

Solution:

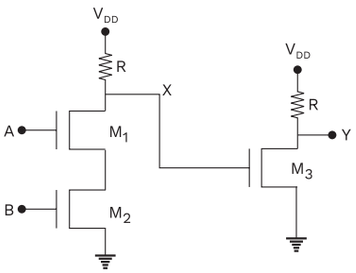


Fig. 8.61

Truth table:

A	B	M_1	M_2	X	M_3	Y
0	0	OFF	OFF	1	ON	0
0	1	OFF	ON	1	ON	0
1	0	ON	OFF	1	ON	0
1	1	ON	ON	0	OFF	1

Table 8.17

So, the output expression, Y = AB.

Example 6:

Find the expression for Y in the given MOSFET circuit.

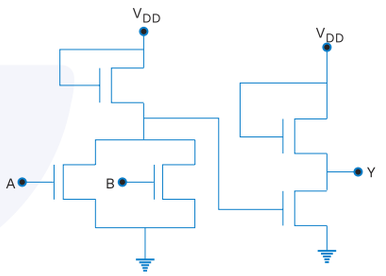


Fig. 8.62

Solution:

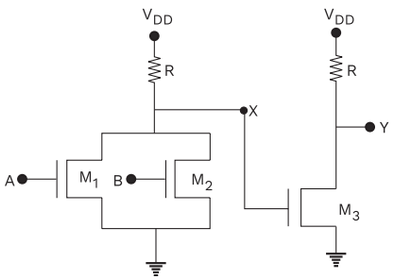


Fig. 8.63

Truth table:

A	B	M_1	M_2	X	M_3	Y
0	0	OFF	OFF	1	ON	0
0	1	OFF	ON	0	OFF	1
1	0	ON	OFF	0	OFF	1
1	1	ON	ON	0	OFF	1

Table 8.18

So, the output expression, Y = A+B.

Example 7:

Find the expression for Y in the given MOSFET circuit.

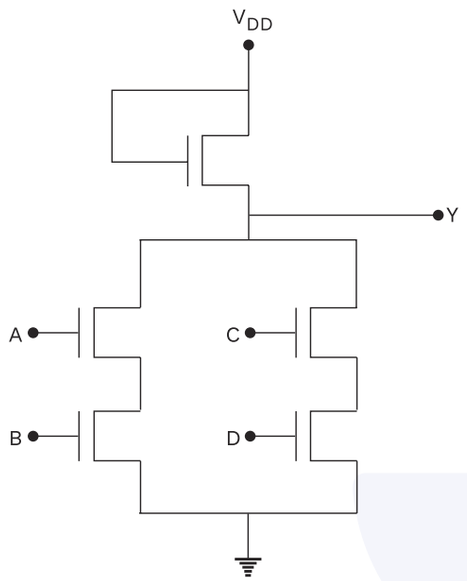


Fig. 8.64

Previous Years' Questions

For the circuit shown in the figure, P and Q are the inputs and Y is the output.

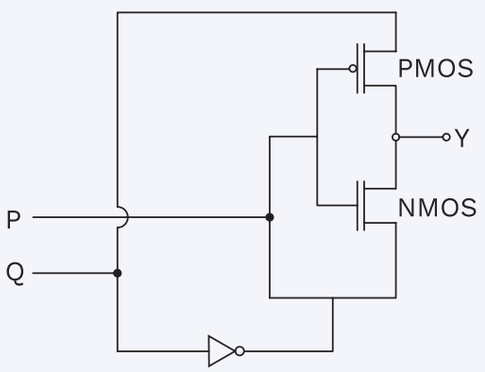


Fig. 8.65

The logic implemented by the circuit is-

(A) XNOR (B) XOR

(C) NOR (D) OR

Solution: (B)

Solution:

Method 1:

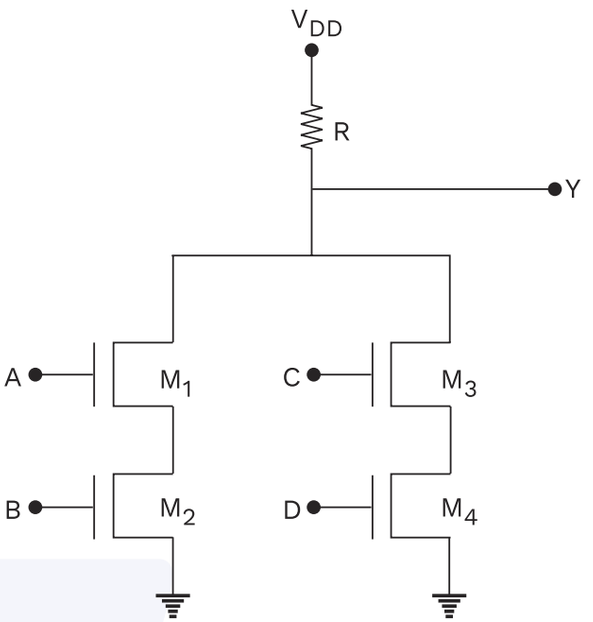


Fig. 8.66

Previous Years' Questions

The full forms of the abbreviations TTL and CMOS in reference to logic families are

(A) Triple Transistor logic and Chip Metal Oxide Semiconductor.

(B) Tristate Transistor Logic and Chip Metal Oxide Semiconductor.

(C) Transistor Transistor Logic and Complementary Metal Oxide Semiconductor.

(D) Tristate Transistor Logic and Complementary Metal Oxide Silicon.

Solution: (C)

Truth table:

A	B	C	D	M_1	M_2	M_3	M_4	Y
0	0	0	0	OFF	OFF	OFF	OFF	1
0	0	0	1	OFF	OFF	OFF	ON	1
0	0	1	0	OFF	OFF	ON	OFF	1
0	0	1	1	OFF	OFF	ON	ON	0
0	1	0	0	OFF	ON	OFF	OFF	1
0	1	0	1	OFF	ON	OFF	ON	1
0	1	1	0	OFF	ON	ON	OFF	1
0	1	1	1	OFF	ON	ON	ON	0
1	0	0	0	ON	OFF	OFF	OFF	1
1	0	0	1	ON	OFF	OFF	ON	1
1	0	1	0	ON	OFF	ON	OFF	1
1	0	1	1	ON	OFF	ON	ON	0
1	1	0	0	ON	ON	OFF	OFF	0
1	1	0	1	ON	ON	OFF	ON	0
1	1	1	0	ON	ON	ON	OFF	0
1	1	1	1	ON	ON	ON	ON	0

Table 8.19

$Y = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 10)$

or $Y = \pi M(3, 7, 11, 12, 13, 14, 15)$

Realisation using K-map

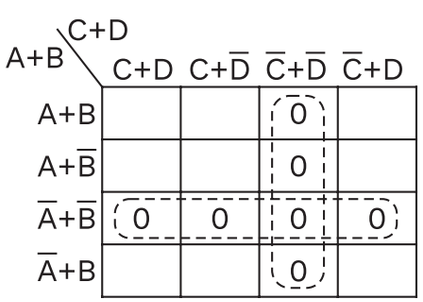


Fig. 8.67

$Y = (\overline{A} + \overline{B}).(\overline{C} + \overline{D})$

Method 2:

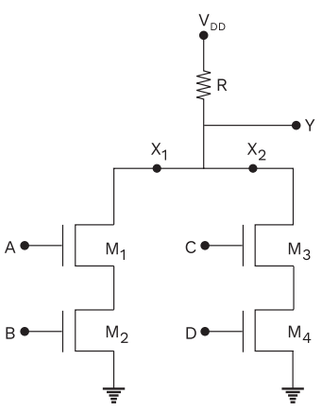


Fig. 8.68

Here, X1 = AB

$$X2 = CD$$

$$Y = \overline{X_1 + X_2}$$

$$Y = \overline{AB + CD}$$

$$Y = \overline{AB}.\overline{CD}$$

$$Y = \left(\overline{A} + \overline{B}\right).\left(\overline{C} + \overline{D}\right)$$

PMOS Logic

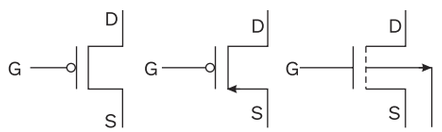


Fig. 8.69 p Channel MOSFET

VGS £ 0 → PMOS is in ON-state → short circuit
VGS > 0 → PMOS is in OFF-state → open circuit

PMOS NOT Gate:

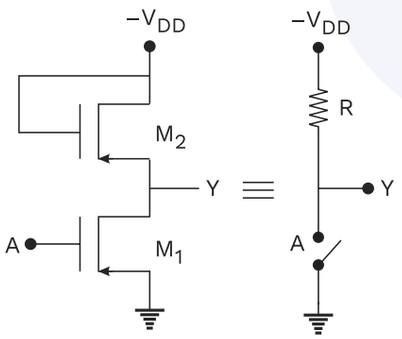


Fig. 8.70 PMOS NOT Gate

- If A = 0, then the MOSFET M_1 will be in ON-state and the output voltage of PMOS will be 0 volt (Y = logic '1', because supply is $-V_{DD}$).
- If A = 1, then the MOSFET M_1 will be in OFF-state and the output voltage of PMOS will be $-V_{DD}$ (Y = logic '0', because supply is $-V_{DD}$).

Truth Table:

A	M_1	Output Voltage	Y
0	ON	0 V	1 (logic '1')
1	OFF	$-V_{DD}$	0 (logic '0')

Table 8.20

PMOS NOR Gate

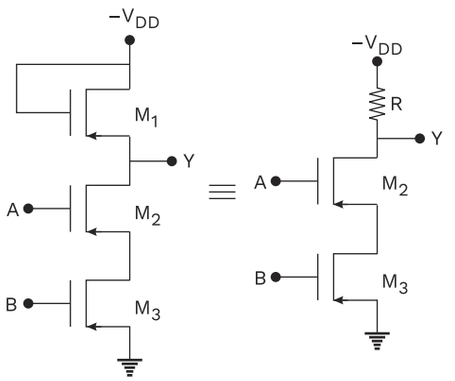


Fig. 8.71 PMOS NOR Gate

- If A = 0 and B = 0, then the MOSFETs M_2 and M_3 will be in ON-state and the output voltage of PMOS circuit will be 0 volt (Y = logic '1').
- If A = 0 and B = 1, then the MOSFET M_2 will be in ON-state and M_3 will be in OFF-state, and the output voltage will be $-V_{DD}$ (Y = logic '0').
- If A = 1 and B = 0, then the MOSFET M_2 will be in OFF-state and M_3 will be ON-state and the output voltage will be $-V_{DD}$ (Y = logic '0').
- If A = 1 and B = 1, v then the MOSFETs M_2 and M_3 will be in OFF-state and the output voltage of PMOS circuit will be $-V_{DD}$ (Y = logic '0').

Truth table:

A	B	Output Voltage	Y
0	0	0 Volt	1 (logic '1')
0	1	$-V_{DD}$	0 (logic '0')
1	0	$-V_{DD}$	0 (logic '0')
1	1	$-V_{DD}$	0 (logic '0')

Table 8.21

Cmos Logic

CMOS logic family uses both PMOS and NMOS in the same circuit to realise several advantages over the PMOS and NMOS families.

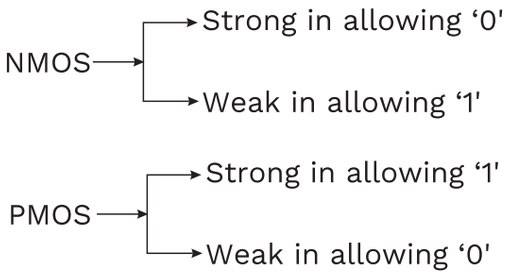


Fig. 8.72

CMOS NOT Gate

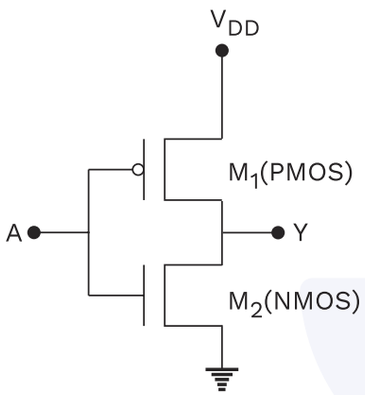


Fig. 8.73 CMOS NOT Gate

- If A = 0, then the MOSFET M_1 (PMOS) will be in ON-state and M_2 (NMOS) will be in OFF-state, and the output will be logic '1' (Y = 1).
- If A = 1, then the MOSFET M_1 (PMOS) will be in OFF-state and M_2 (NMOS) will be in ON-state and the output will be logic '0' (Y = 0).

Truth table:

A	M_1 (PMOS)	M_2 (NMOS)	Y
0	ON	OFF	1
1	OFF	ON	0

Table 8.22

CMOS NAND Gate

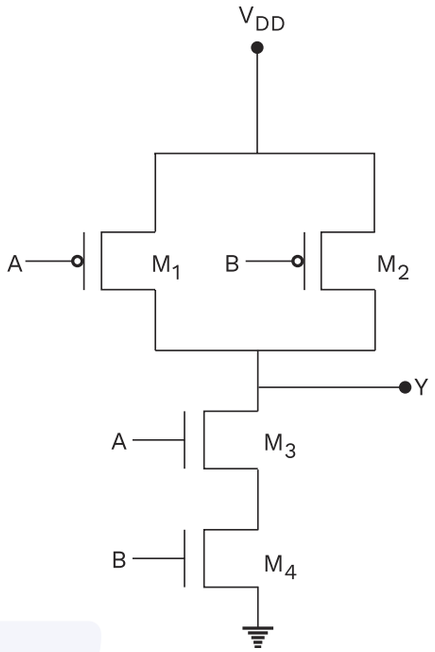


Fig. 8.74 CMOS NAND Gate

- If A = 0 and B = 0, then the MOSFETs M_1 and M_2 will be in ON-state, MOSFETs M_3 and M_4 will be in OFF-state and the output will be logic '1' (Y=1).
- If A = 0 and B = 1, then the MOSFETs M_1 and M_4 will be in ON-state, MOSFETs M_2 and M_3 will be in OFF-state and the output will be logic '1' (Y=1).
- If A = 1 and B = 0, then the MOSFETs M_1 and M_4 will be in OFF-state, MOSFETs M_2 and M_3 will be ON-state and the output will be logic '1' (Y=1).
- If A = 1 and B = 1, then the MOSFETs M_1 and M_2 will be in OFF-state, MOSFETs M_3 and M_4 will be in ON-state and the output will be logic '0'. (Y = 0).

Truth table:

A	B	M_1	M_2	M_3	M_4	Y
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

Table 8.23

CMOS NOR Gate

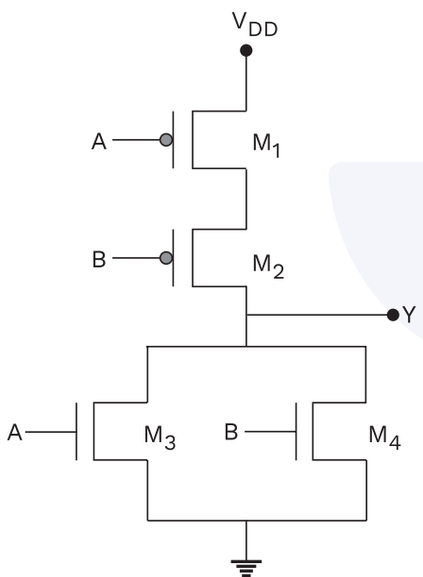


Fig. 8.75 CMOS NOR Gate

- If A = 0 and B = 0, then the MOSFETs M_1 and M_2 will be in ON-state, MOSFETs M_3 and M_4 will be in OFF-state and the output will be logic '1' (Y = 1).
- If A = 0 and B = 1, then the MOSFETs M_1 and M_4 will be in ON-state, MOSFETs M_2 and M_3 will be in OFF-state and the output will be logic '0' (Y = 0).
- If A = 1 and B = 0, then the MOSFETs M_1 and M_4 will be in OFF-state, MOSFETs M_2 and M_3 will be in ON-state and the output will be logic '0' (Y = 0).
- If A = 1 and B = 1, then the MOSFETs M_1 and M_2 will be in OFF-state, MOSFETs M_3 and M_4 will be in ON-state and the output will be logic '0' (Y = 0).

Truth table:

A	B	M_1	M_2	M_3	M_4	Y
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	0
1	0	OFF	ON	ON	OFF	0
1	1	OFF	OFF	ON	ON	0

Table 8.24

CMOS transmission gate:

- CMOS transmission gate consists of one NMOS and one PMOS transistor, connected in parallel.
- It is a bilateral switch which can be controlled by externally applied logic levels.

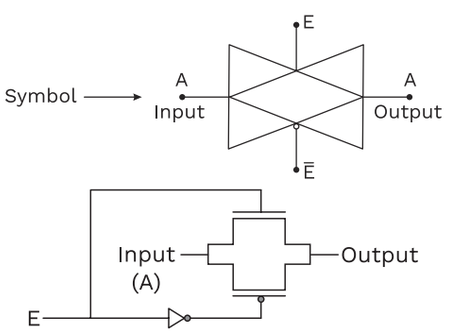


Fig. 8.76 CMOS Symbol

- In a transmission gate, if enable E = 0, then NMOS and PMOS both will be in OFF-state. So, irrespective of inputs (logic '0' or logic '1'), the output of the circuit will be in high-impedance state.

- If enable E = 1 and A = 0, then NMOS and PMOS both will be in ON-state. So, the output of the circuit will be the same as the input (works as a short circuit) means Y = 0.
- If enable E = 1 and A = 1, then NMOS and PMOS both will be in ON-state. So, the output of the circuit will be Y=1.

Truth table:

E	A	Y
0	X	High-impedance state
1	0	0
1	1	1

Table 8.25

- **2×1 MUX using transmission gate:**

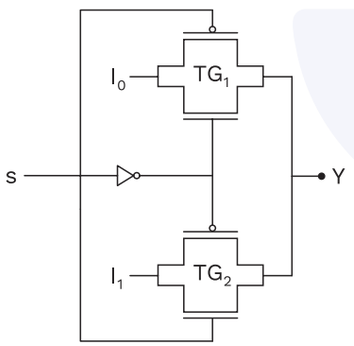


Fig. 8.77 2×1 MUX Using Transmission Gate

- If select line S=0, then transmission gate-1 (TG_1) will be in ON-state and TG_2 will be in OFF-state. So, the output of the 2×1 MUX will be I_0(Y=I_0).
- If select line S=1, then transmission gate-2 (TG_2) will be in ON-state and TG_1 will be in OFF-state. So, the output of the 2×1 MUX will be I_1(Y=I_1).

Truth table:

S	Y
0	I_0
1	I_1

Table 8.26

We know that using a multiplexer, we can implement different gates (AND gate, OR gate, NOR gate, NAND gate etc,). That is why a multiplexer is called as a universal logic gate. So, we can say that by using CMOS transmission gate we can implement any logic gate.

Chapter Summary

- In logic families propagation delay, power dissipation and figure of merit should be as minimum as possible.
- Fan-out and noise margin should be as maximum as possible.
- Current source: When logic gate output is '1' (transistor off), then the driving gate supplies current to the load gate at a high state.
- Current sink: When logic gate output is '0' (transistor ON), then the driving gate receives current from the load gate in a low state.
- HTL is the modified form of DTL.
- RTL and DTL have low speed, low fan-out and high power dissipation.
- HTL has an excellent noise margin and the largest voltage swing.
- ECL is the fastest in the logic families.
- Any floating input in the TTL circuit is considered as logic '1'.
- DTL provides wired AND logic.

- DTL provides wired AND logic.
- ECL provides wired OR logic.
- PMOS is called as 'pull up' network, and NMOS is called as 'pull down' network.
- CMOS is the slowest logic familiy.
- CMOS is also used in a monostable multivibrator.
- **Comparison of logic families**

Parameter	TTL	ECL	NMOS	CMOS
t_{pd} (ns)	10	1	250	70
P_{diss}(mW)	10	55	1	0.01
FOM(pJ)	100	55	250	0.7
NM(V)	0.4	0.3	1.5	$V_{DD}/2$
Fan-out	10	25	5	50

Table 8.27

Practice Questions

1. The open collector of the gates is connected together as shown in the given figure. The logic expression for Y will be:

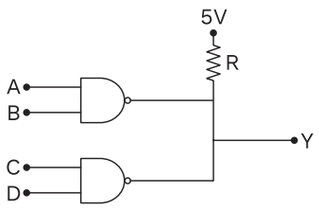


Fig. 8.78

(A) $(A + B)(C + D)$

(B) $\overline{(A+B)}+\overline{(C+D)}$

(C) $\overline{AB+CD}$

(D) $\overline{A+B+C+D}$

2. 'ECL' has very high switching speed because the transistors are:

(A) Switching between active and saturation region

(B) Switching between cut-off and active region

(C) Switching between cut-off and saturation region

(D) None of the above

3. Consider the following statements:

I. TTL has high switching speed and good fan-out capability

II. ECL has least propagation delay

III. I2L uses multi-collector transistor

IV. NMOS has more silicon area

Which of the above statements are correct?

(A) I, II and III

(B) II and IV

(C) I, III and IV

(D) I, II, III and IV

4. The circuit with inputs A and B shown in the given figure performs:

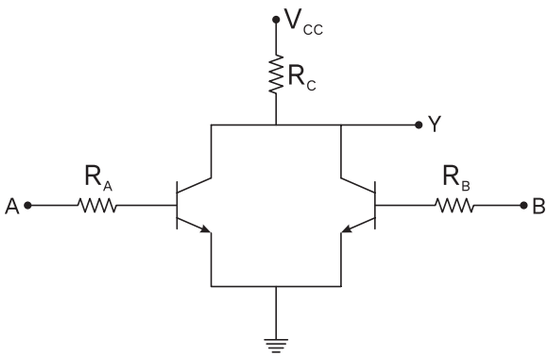


Fig. 8.79

(A) AND Operation
(B) NAND Operation
(C) OR Operation
(D) NOR Operation

5. A logic family has threshold voltage VR = 1V, minimum guaranteed output high voltage VOH = 4V, minimum accepted input high voltage VIH = 2V, maximum guaranteed output low voltage VOL = 0.5 and maximum accepted input low voltage VIL = 1.5 V. Its noise margin is _____ volt.

6. Find the expression for Y in the given MOSFET circuit.

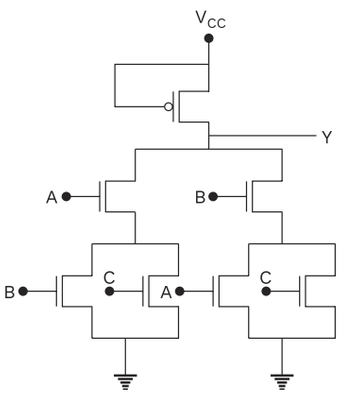


Fig. 8.80

(A) $Y = (\overline{A} + \overline{B})(\overline{A} + \overline{C})(\overline{B} + \overline{C})$

(B) $Y = (B + C).A + (A + C)B$

(C) $Y = \overline{(A + BC)(B + AC)}$

(D) $Y = \overline{(AB + C)(AC + B)}$

7. Match List 1 (logic gates) with List 2 (operations) and select the correct answer using the codes given below:

List 1	List 2
A. TTL	I. More logical swing
B. ECL	II. Low power dissipation
C. DCTL	III. Current hogging
D. CMOS	IV. NOR/OR output
	V. Totem-pole output

Codes:

	A	B	C	D
(A)	III	II	V	I
(B)	III	II	IV	V
(C)	II	III	IV	V
(D)	V	IV	III	II

Practice Solutions

1. Solution: (C) **2. Solution: (B)** **3. Solution: (B)** **4. Solution: (D)**

5. Solution: 1 **6. Solution: (A)** **7. Solution: (D)**